Five Solidity questions, answered

Solidity 0.8.x · EVM · Interview bank

SOL–F–04 Foundation Knowledge

What is the purpose of `view` and `pure` functions?

Answer summary · authored

Off-chain eth_call has no network transaction fee. The same computation inside a transaction uses gas.

Persistent, writable temporary, read-only

SOL-F-01 Foundation Knowledge

What is the difference between `storage`, `memory`, and `calldata` in Solidity?

Answer summary · authored

storage persists between calls. memory is writable scratch space for a call. calldata is read-only argument data. Use calldata for reference-type inputs you do not need to modify.

What _; does in a modifier

SOL-F-02 Foundation Knowledge

What is a function modifier and can you provide a simple example?

Illustrative modifier fragment

```
modifier onlyOwner() {
  require(msg.sender == owner);
  _;
}
```

The caller check runs before the function body. _; runs the body; code after it runs afterward.

The example assumes owner is defined and initialized; it is a modifier fragment, not a complete ownership contract.

Read-only does not mean zero computation

SOL-F-04 Foundation Knowledge

What is the purpose of `view` and `pure` functions?

Answer summary · authored

view may read state but cannot modify it. pure may neither read nor modify state. An off-chain eth_call pays no network fee; execution inside a transaction still consumes gas.

New logic, same proxy state

SOL-I-02 | Intermediate | Architecture

What is an upgradeability proxy pattern, and how does `delegatecall` enable it?

Answer summary · authored

Implementation code uses the proxy's storage and address, preserving msg.sender and msg.value. Upgrades change the implementation pointer; storage layout must stay compatible.

Storage packing: 2 or 3 slots

SOL-I-03 | Intermediate | Gas Optimization

When packing structs for gas efficiency, how should you order the variables? Give an example.

Authored example · Each box is 32 bytes

2 slots	2 × uint128	uint256	
3 slots	uint128	uint256	uint128

Fewer slots do not guarantee cheaper writes; gas depends on access patterns.

Do not reorder existing fields in a deployed proxy implementation's storage layout.
